

CBSE Class 11 Physics — Exam Cheat Sheet

Rationalised NCERT (2023–24 onward) · Ch 1–7 (Gravitation up to variation of g) · ★ = high-priority derivation

Ch 1 · Units and Measurements

FORMULAS

► Dimensional formulae (learn this table cold)

Quantity	Dimensions	SI unit
Force, Weight, Tension	$[MLT^{-2}]$	N
Work, Energy, Torque	$[ML^2T^{-2}]$	J; N m
Power	$[ML^2T^{-3}]$	W
Pressure, Stress, Elastic modulus	$[ML^{-1}T^{-2}]$	Pa
Momentum, Impulse	$[MLT^{-1}]$	kg m s ⁻¹ ; N s
Angular momentum, Planck's h	$[ML^2T^{-1}]$	J s
Moment of inertia I	$[ML^2]$	kg m ²
Gravitational constant G	$[M^{-1}L^3T^{-2}]$	N m ² kg ⁻²
Surface tension, Spring constant k	$[MT^{-2}]$	N m ⁻¹
Coefficient of viscosity η	$[ML^{-1}T^{-1}]$	Pa s (CGS: poise)
Frequency ν , ω , Velocity gradient	$[T^{-1}]$	Hz; rad s ⁻¹
Acceleration, g , Gravitational field	$[LT^{-2}]$	m s ⁻² ; N kg ⁻¹
Density	$[ML^{-3}]$	kg m ⁻³
Angle, Strain, μ , e , Refractive index	$[M^0L^0T^0]$	rad / none

Same dimensions $\Rightarrow$ may be added/equated. Work, torque and energy share $[ML^2T^{-2}]$ yet differ physically.

► Significant figures

- **Counting:** non-zero digits, and zeros *between* them, count; *leading* zeros never ($0.00250 \Rightarrow 3$); trailing zeros only with a decimal point ($2.500 \Rightarrow 4$; 2500 ambiguous $\rightarrow 2.5 \times 10^3$).
- **Add / subtract:** keep the *fewest decimal places*. $436.32 + 227.2 + 0.301 = 663.8$
- **Multiply / divide:** keep the *fewest significant figures*. $4.237 \times 2.51 = 10.6$
- **Rounding:** dropped digit > 5 up, < 5 down, exactly 5 $\Rightarrow$ make the preceding digit **even** ($2.745 \rightarrow 2.74$, $2.735 \rightarrow 2.74$).
- **Exact numbers** (counts, the 2 in $2\pi r$) have infinite s.f. — they never limit the answer.

► Unit conversion

- $n_1 u_1 = n_2 u_2$ — magnitude $\times$ unit is invariant: bigger unit $\Rightarrow$ smaller number
- $n_2 = n_1 \left[\frac{M_1}{M_2} \right]^a \left[\frac{L_1}{L_2} \right]^b \left[\frac{T_1}{T_2} \right]^c$ — a, b, c = exponents in its dimensional formula
- $1 \text{ N} = 10^5 \text{ dyne}$; $1 \text{ J} = 10^7 \text{ erg}$; $1 \text{ Pa} = 10 \text{ dyne cm}^{-2}$; $1 \text{ kg m}^2 \text{ s}^{-1} = 10^7 \text{ CGS units}$

DERIVATIONS

★ Time period of a simple pendulum by the dimensional method

1. Assume $T = k m^a l^b g^c$, k dimensionless.
 2. Write dimensions: $[M^0 L^0 T^1] = [M]^a [L]^b [L T^{-2}]^c$.
 3. Collect: $[M^0 L^0 T^1] = [M^a L^{b+c} T^{-2c}]$.
 4. Equate exponents: $a = 0$; $b + c = 0$; $-2c = 1$.
 5. $c = -\frac{1}{2}$, $b = +\frac{1}{2}$, $a = 0$ — **T is independent of mass.**
- $T = k\sqrt{l/g}$; **experiment gives** $k = 2\pi \Rightarrow T = 2\pi\sqrt{l/g}$.

Checking homogeneity of $v^2 = u^2 + 2as$

1. Every additive term in a correct equation has identical dimensions.
2. LHS: $[v^2] = [L T^{-1}]^2 = [L^2 T^{-2}]$.
3. 1st term RHS: $[u^2] = [L^2 T^{-2}]$.
4. 2nd term RHS: $[2as] = [L T^{-2}][L] = [L^2 T^{-2}]$.
5. All three match $\Rightarrow$ dimensionally homogeneous, hence dimensionally correct.

Necessary, not sufficient: the factor 2 is unchecked, so $v^2 = u^2 + 3as$ also "passes".

STANDARD RESULTS

► Limitations of dimensional analysis

- Cannot find **dimensionless constants** (k , 2π , $\frac{1}{2}$) — only the *form* of the relation.
- Fails for quantities depending on **more than three** variables (only 3 equations from M, L, T).
- Cannot handle **trigonometric, exponential or logarithmic** relations ($\sin \theta$, e^{-kt}).
- Cannot derive relations containing **sums of terms**, e.g. $s = ut + \frac{1}{2}at^2$.
- Cannot **distinguish** quantities with identical dimensions (work vs torque; energy vs moment of force).
- Says nothing about **scalar vs vector**, or whether an equation is physically meaningful.

► Useful constants

- $g = 9.8 \text{ m s}^{-2}$, $R_E = 6.4 \times 10^6 \text{ m}$, $M_E \approx 6 \times 10^{24} \text{ kg}$, $G = 6.67 \times 10^{-11} \text{ SI}$, $1 \text{ hp} = 746 \text{ W}$, $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$.

Only if covered in class — Error propagation

- $\bar{a} = \frac{1}{n} \sum a_i$; mean absolute error $\Delta \bar{a} = \frac{1}{n} \sum |\Delta a_i|$; relative $\Delta \bar{a} / \bar{a}$; percentage $\times 100\%$.
- **Sum or difference** $Z = A \pm B$: $\Delta Z = \Delta A + \Delta B$ — *absolute errors add*, even for a difference.
- **Product or quotient** $Z = AB$ or A/B : $\frac{\Delta Z}{Z} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$ — *relative errors add*.
- **Powers** $Z = \frac{A^p B^q}{C^r}$: $\frac{\Delta Z}{Z} = p \frac{\Delta A}{A} + q \frac{\Delta B}{B} + r \frac{\Delta C}{C}$ — the exponent *multiplies* the error, so measure high powers most carefully.

Traps — Ch 1

1. $+$, $- \Rightarrow$ count **decimal places**; $\times$, $\div \Rightarrow$ count **significant figures**. Swapping them is the classic slip.
2. Errors **always add**. For $Z = A - B$ you still write $\Delta Z = \Delta A + \Delta B$ — never subtract errors.
3. Dimensionally correct $\neq$ physically correct: dimensional analysis cannot detect a wrong numerical factor, nor tell torque from work.
4. Arguments of $\sin$, $\cos$, $\log$ and exponents must be **dimensionless** — use this to find an unknown constant's dimensions.
5. In $n_2 = n_1 [M_1/M_2]^a \dots$: 1 = **old** system, 2 = **new**. Inverting the ratios is a guaranteed blunder.

Ch 2 · Motion in a Straight Line

FORMULAS

► Definitions

- $\bar{v} = \frac{\Delta x}{\Delta t}$ — average velocity; displacement per unit time; vector [**m s⁻¹**]
- Average speed = $\frac{\text{path length}}{\text{time}}$ — scalar; $\geq |\bar{v}|$ [**m s⁻¹**]
- $v = \frac{dx}{dt}$, $a = \frac{dv}{dt} = \frac{d^2x}{dt^2}$ — instantaneous velocity and acceleration [**m s⁻¹**, **m s⁻²**]
- $a = v \frac{dv}{dx}$ — use when a is given as a function of position x , not of time
- Equal *distances* at v_1, v_2 : $\bar{v} = \frac{2v_1 v_2}{v_1 + v_2}$; equal *times*: $\bar{v} = \frac{v_1 + v_2}{2}$.

► Uniformly accelerated motion (constant a only)

- $v = u + at$ — velocity–time; no s
- $s = ut + \frac{1}{2}at^2$ — displacement–time; no v [m]
- $v^2 = u^2 + 2as$ — no t ; use when time is neither given nor asked
- $s = \left(\frac{u+v}{2}\right)t$ — no a
- $s_{n^{\text{th}}} = u + \frac{a}{2}(2n-1)$ — displacement **during** the n^{th} second [m, not m s⁻¹]

► Free fall & stopping

- Dropped from rest (down positive, $u = 0$): $v = gt$, $h = \frac{1}{2}gt^2$, $v = \sqrt{2gh}$
- Thrown up with u ($a = -g$): $t_{\text{up}} = u/g$, $H_{\text{max}} = u^2/2g$, $T = 2u/g$; returns with speed u .
- Stopping distance $s = \frac{u^2}{2a}$ (a = retardation) — $\propto u^2$: double the speed $\Rightarrow 4\times$ the distance
- Relative velocity 1D: $v_{AB} = v_A - v_B$ (A as seen by B); same direction $\Rightarrow$ subtract, opposite $\Rightarrow$ add magnitudes.

DERIVATIONS

★ Three equations of motion — graphical (v - t graph)

- v - t graph, constant a : line AB from $(0, u)$ to (t, v) ; $BC \perp t$ -axis, $AD \parallel t$ -axis $\Rightarrow BD = v - u$, $AD = OC = t$.
- Eq. 1:** slope = $a = \frac{BD}{AD} = \frac{v-u}{t} \Rightarrow v = u + at$.
- Eq. 2:** displacement = area $OABC$ = rectangle + triangle = $ut + \frac{1}{2}t(at) \Rightarrow s = ut + \frac{1}{2}at^2$.
- Eq. 3:** same area as a trapezium = $\frac{1}{2}(u+v)t$; put $t = (v-u)/a$ from Eq. 1.
- $s = \frac{(u+v)(v-u)}{2a} = \frac{v^2-u^2}{2a} \Rightarrow v^2 = u^2 + 2as$.

★ Three equations of motion — calculus

- $a = \frac{dv}{dt}$ with a constant $\Rightarrow dv = a dt$.
- $\int_u^v dv = a \int_0^t dt \Rightarrow v - u = at$. (1)
- $\frac{dx}{dt} = u + at \Rightarrow \int_{x_0}^x dx = \int_0^t (u + at) dt$.
- $x - x_0 = s = ut + \frac{1}{2}at^2$. (2)
- Then $a = \frac{dv}{dt} = \frac{dv}{dx} \frac{dx}{dt} = v \frac{dv}{dx} \Rightarrow v dv = a dx$.
- $\int_u^v v dv = a \int_0^s dx \Rightarrow \frac{v^2-u^2}{2} = as \Rightarrow v^2 = u^2 + 2as$. (3)

Distance travelled in the n^{th} second

- $s_{n^{\text{th}}} = s_n - s_{n-1}$ (distance in n s minus in $n-1$ s).
- $s_n = un + \frac{1}{2}an^2$; $s_{n-1} = u(n-1) + \frac{1}{2}a(n-1)^2$.
- Subtract: $s_{n^{\text{th}}} = u + \frac{a}{2}[n^2 - (n-1)^2] = u + \frac{a}{2}(2n-1)$.

Unit: metre — a displacement over a 1 s interval, numerically equal to that second's average velocity.

STANDARD RESULTS — GRAPH INTERPRETATION

Graph	Slope gives	Area gives
x - t	velocity v	no physical meaning
v - t	acceleration a	displacement (signed)
a - t	jerk (not in syllabus)	change in velocity Δv

Motion	x - t shape	v - t shape
At rest	horizontal line	on the t -axis ($v = 0$)
Uniform velocity	straight line, slope = v	horizontal line
Uniform acceleration ($u = 0$)	parabola opening upward	straight line through origin
Uniform retardation	parabola, slope decreasing	straight line, negative slope

- An x - t graph is never **vertical** (infinite velocity) and never doubles back (one position per instant).
- Area below the t -axis of a v - t graph is negative: add magnitudes for *distance*, add signed for *displacement*.

Traps — Ch 2

- $s_{n^{\text{th}}} = u + \frac{a}{2}(2n-1)$ has unit **metre**, though its form looks like a velocity. Do not write m s⁻¹.

- Fix the + direction **once**. Thrown up with "up positive" $\Rightarrow a = -g$ throughout, including at the top and on the way down.
- At the highest point $v = 0$ but $a = g \neq 0$. "Velocity zero" never means "acceleration zero".
- The three equations need **constant** a . If a varies, integrate — use $a = v dv/dx$ when $a = a(x)$.
- Negative acceleration $\neq$ deceleration. Motion slows only when v and a have **opposite signs**.

Ch 3 · Motion in a Plane

FORMULAS

► Vector addition & resolution

- Parallelogram law** (θ = angle *between* them, tails together): $R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$, $\tan \alpha = \frac{B \sin \theta}{A + B \cos \theta}$ — α from $\vec{A}$
- Limits: $0^\circ \Rightarrow A + B$ (max); $90^\circ \Rightarrow \sqrt{A^2 + B^2}$; $180^\circ \Rightarrow |A - B|$ (min).
- Triangle law:** tail of $\vec{B}$ at head of $\vec{A}$; the closing side is $\vec{R}$.
- Polygon law:** n vectors; closed $\Rightarrow \vec{R} = 0$.
- Subtraction:** $\vec{A} - \vec{B} = \vec{A} + (-\vec{B})$, so $|\vec{A} - \vec{B}| = \sqrt{A^2 + B^2 - 2AB \cos \theta}$.
- Resolution** (θ from the $+x$ axis): $A_x = A \cos \theta$, $A_y = A \sin \theta$; $A = \sqrt{A_x^2 + A_y^2}$, $\tan \theta = A_y/A_x$.
- Unit vector** $\hat{A} = \vec{A}/|\vec{A}|$ (dimensionless, magnitude 1); $\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$.

► Products

- Dot:** $\vec{A} \cdot \vec{B} = AB \cos \theta = A_x B_x + A_y B_y + A_z B_z$; commutative; $\hat{i} \cdot \hat{i} = 1$, $\hat{i} \cdot \hat{j} = 0$; zero $\Leftarrow \perp$; gives W .
- Cross:** $|\vec{A} \times \vec{B}| = AB \sin \theta$, $\perp$ both (right hand); $= -\vec{B} \times \vec{A}$; $\vec{A} \times \vec{A} = 0$; $\hat{i} \times \hat{j} = \hat{k}$, $\hat{j} \times \hat{k} = \hat{i}$, $\hat{k} \times \hat{i} = \hat{j}$; zero $\Leftarrow$ parallel; gives $\vec{\tau}$.
- $\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$; $\cos \theta = \frac{\vec{A} \cdot \vec{B}}{AB}$.

► Constant acceleration in a plane (each component independent)

- $\vec{v} = \vec{u} + \vec{a}t$; $\vec{r} = \vec{r}_0 + \vec{u}t + \frac{1}{2}\vec{a}t^2$; $\vec{v} \cdot \vec{v} = \vec{u} \cdot \vec{u} + 2\vec{a} \cdot (\vec{r} - \vec{r}_0)$
- Components: $v_x = u_x + a_x t$, $x = u_x t + \frac{1}{2}a_x t^2$; same independently in y .

► Projectile motion (u at angle θ to the horizontal, same landing level)

- $T = \frac{2u \sin \theta}{g}$ — time of flight [s]; time to peak = $T/2$
- $H = \frac{u^2 \sin^2 \theta}{2g}$ — maximum height [m]
- $R = \frac{u^2 \sin 2\theta}{g}$ — horizontal range on level ground [m]; $R_{\text{max}} = u^2/g$ at $\theta = 45^\circ$
- $y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$ — trajectory: a parabola; $R = 4H \cot \theta$

- At t : $v_x = u \cos \theta$, $v_y = u \sin \theta - gt$, $v = \sqrt{v_x^2 + v_y^2}$, $\tan \beta = v_y/v_x$. Top: speed $u \cos \theta$, $a = g$ down.

- Horizontal projection from h :** $t = \sqrt{2h/g}$, $x = u\sqrt{2h/g}$, $v_y = \sqrt{2gh}$, $v = \sqrt{u^2 + 2gh}$, $\tan \beta = \sqrt{2gh}/u$ below horizontal, $y = \frac{gx^2}{2u^2}$ (y down).

► Uniform circular motion

- $v = \omega r$; $\omega = \frac{2\pi}{T} = 2\pi\nu$ — angular velocity [rad s⁻¹]
- $a_c = \frac{v^2}{r} = \omega^2 r = \frac{4\pi^2 r}{T^2}$ — centripetal acceleration, directed to the centre [m s⁻²]
- Non-uniform: $a_t = dv/dt$, net $a = \sqrt{a_c^2 + a_t^2}$; constant speed $\Rightarrow a_t = 0$.

► Relative velocity in 2D

- $\vec{v}_{AB} = \vec{v}_A - \vec{v}_B$ — velocity of A relative to B ; $\vec{v}_{AB} = -\vec{v}_{BA}$

- **Rain-umbrella:** rain v_r down, man v_m across $\Rightarrow$ tilt **forward** at $\tan \theta = v_m/v_r$ from the vertical; apparent speed $\sqrt{v_r^2 + v_m^2}$.
- **River-boat, shortest time:** steer straight across $\Rightarrow t = d/v_b$, drift $v_r d/v_b$, resultant $\sqrt{v_b^2 + v_r^2}$.
- **Zero drift:** steer upstream at $\sin \theta = v_r/v_b$ from the normal $\Rightarrow$ speed $\sqrt{v_b^2 - v_r^2}$, $t = d/\sqrt{v_b^2 - v_r^2}$; needs $v_b > v_r$.

DERIVATIONS

★ Parallelogram law — magnitude and direction of the resultant

1. $\vec{A} = \vec{OP}$, $\vec{B} = \vec{OQ}$, angle θ between; complete parallelogram $OPSQ \Rightarrow \vec{R} = \vec{OS}$, $PS = B$.
2. Drop $SN \perp OP$ produced $\Rightarrow PN = B \cos \theta$, $SN = B \sin \theta$.
3. In $\triangle ONS$: $R^2 = ON^2 + SN^2 = (A + B \cos \theta)^2 + (B \sin \theta)^2$.
4. Expand: $R^2 = A^2 + 2AB \cos \theta + B^2(\cos^2 \theta + \sin^2 \theta)$.
5. $\Rightarrow R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$.
6. $\tan \alpha = \frac{SN}{ON} = \frac{B \sin \theta}{A + B \cos \theta}$, α from $\vec{A}$.

★ Projectile: trajectory is a parabola, T , H , R

1. Origin at launch, x horizontal, y up: $a_x = 0$, $a_y = -g$, $u_x = u \cos \theta$, $u_y = u \sin \theta$; the two motions are **independent**.
2. Horizontal: $x = (u \cos \theta)t \Rightarrow t = \frac{x}{u \cos \theta}$.
3. Vertical: $y = (u \sin \theta)t - \frac{1}{2}gt^2$; substitute t from step 2:
4. $y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$ — form $y = ax - bx^2 \Rightarrow$ the path is a **parabola**.
5. T : put $y = 0$: $t(u \sin \theta - \frac{1}{2}gt) = 0 \Rightarrow T = \frac{2u \sin \theta}{g}$.
6. H : at the top $v_y = 0$; $0 = u^2 \sin^2 \theta - 2gH \Rightarrow H = \frac{u^2 \sin^2 \theta}{2g}$.
7. R : $R = (u \cos \theta)T = \frac{2u^2 \sin \theta \cos \theta}{g} = \frac{u^2 \sin 2\theta}{g}$.

R max at $\sin 2\theta = 1$, i.e. $\theta = 45^\circ$, $R_{\max} = u^2/g$. Since $\sin 2(90^\circ - \theta) = \sin 2\theta$, **complementary angles** (30° , 60°) give equal R , different T , H .

★ Centripetal acceleration $a_c = v^2/r = \omega^2 r$

1. Particle on a circle of radius r at constant speed v : position vectors $\vec{r}_1, \vec{r}_2$ at t and $t + \Delta t$ subtend $\Delta \theta$ at the centre.
2. Velocity is tangential, so $\vec{v}_1, \vec{v}_2$ (each of length v) turn through the **same** $\Delta \theta$.
3. Position triangle ($\vec{r}_1, \vec{r}_2, \Delta \vec{r}$) and velocity triangle ($\vec{v}_1, \vec{v}_2, \Delta \vec{v}$) are isosceles with the same apex angle $\Rightarrow$ **similar**.
4. Hence $\frac{|\Delta \vec{v}|}{v} = \frac{|\Delta \vec{r}|}{r}$, i.e. $|\Delta \vec{v}| = \frac{v}{r} |\Delta \vec{r}|$.
5. Divide by Δt and let $\Delta t \rightarrow 0$, where $\frac{|\Delta \vec{r}|}{\Delta t} \rightarrow v$:
6. $a = \frac{v^2}{r}$; and $\Delta \vec{v}$ becomes $\perp \vec{v}$, pointing to the centre $\Rightarrow$ **centripetal**.

With $v = \omega r$: $a_c = \frac{v^2}{r} = \omega^2 r = v\omega = \frac{4\pi^2 r}{T^2}$. **Speed constant, velocity not — hence non-zero a .**

Traps — Ch 3

1. Parallelogram: θ is **tail-to-tail**. Triangle: **head-to-tail**, so the figure shows $180^\circ - \theta$. Mixing them flips the sign of $2AB \cos \theta$.
2. $A \cos \theta$ is the x -component only if θ is measured **from the x -axis**. From the y -axis or an incline, sine and cosine swap.
3. In a projectile, $a = g$ downward at **every** point, including the top where $v_y = 0$; v_x never changes.
4. R , T , H assume launch and landing at the **same height**. From a cliff or tower, go back to the component

equations.

5. $a_c = v^2/r$ is an **acceleration**, not a force. Never add a "centrifugal force" in an inertial frame — the inward force is tension, friction, gravity or N .

Ch 4 · Laws of Motion

FORMULAS

► Newton's laws, momentum, impulse

- **1st law (inertia):** a body stays at rest or in uniform motion unless a net external force acts. Defines force qualitatively, and the inertial frame.
- **2nd law:** $\vec{F} = \frac{d\vec{p}}{dt} = m\vec{a}$ (constant m) — gives force quantitatively [**N = kg m s⁻²**]
- $\vec{p} = m\vec{v}$ — linear momentum [**kg m s⁻¹**]
- **3rd law:** $\vec{F}_{AB} = -\vec{F}_{BA}$ — equal, opposite, simultaneous, on **two different** bodies
- Impulse $\vec{J} = \vec{F}_{\text{avg}} \Delta t = \Delta \vec{p} = \int \vec{F} dt =$ area under $F-t$ [**N s**]
- **Conservation:** $\vec{F}_{\text{ext}} = 0 \Rightarrow \vec{p}_{\text{tot}}$ constant: $m_1 \vec{u}_1 + m_2 \vec{u}_2 = m_1 \vec{v}_1 + m_2 \vec{v}_2$.

► Equilibrium of concurrent forces

- $\sum \vec{F} = 0 \Rightarrow \sum F_x = 0$, $\sum F_y = 0$ — necessary and sufficient for a particle
- Three concurrent forces in equilibrium close into a triangle. **Lami:** $\frac{F_1}{\sin \alpha} = \frac{F_2}{\sin \beta} = \frac{F_3}{\sin \gamma}$, each angle being the one **opposite** its force.

► Friction

- $f_s \leq \mu_s N$, $(f_s)_{\max} = \mu_s N$ — **self-adjusting**: equals the applied force up to the limit
- $f_k = \mu_k N$, $\mu_k < \mu_s$ — constant; independent of speed and contact area
- **Angle of friction** λ : $\tan \lambda = \mu_s$ — between the resultant of N , $f_{\max}$ and N
- **Angle of repose** α : $\tan \alpha = \mu_s$ — incline at which sliding just begins; $\alpha = \lambda$
- Rolling friction $\ll$ sliding friction — hence wheels and ball bearings.

► Circular motion dynamics

- $F_c = \frac{mv^2}{r} = m\omega^2 r$ — the *name* of the net inward force, not a new force [**N**]
- Level road: $v_{\max} = \sqrt{\mu_s Rg}$ — friction alone supplies F_c
- Banked, frictionless: $v_0 = \sqrt{Rg \tan \theta}$ — optimum speed; no friction needed
- Banked + friction: $v_{\max} = \sqrt{Rg \frac{\mu_s + \tan \theta}{1 - \mu_s \tan \theta}}$; $v_{\min} = \sqrt{Rg \frac{\tan \theta - \mu_s}{1 + \mu_s \tan \theta}}$ if $\tan \theta > \mu_s$

► Standard systems

- **Lift (N = apparent weight):** rest: mg ; up with a : $m(g + a)$; down: $m(g - a)$; free fall: 0 ; $a > g$ down $\Rightarrow$ presses on the roof.
- **Atwood** ($m_1 > m_2$, light string, ideal pulley): $a = \frac{(m_1 - m_2)g}{m_1 + m_2}$, $T = \frac{2m_1 m_2 g}{m_1 + m_2}$, pulley load = $2T$.
- **Incline θ :** $N = mg \cos \theta$; smooth $a = g \sin \theta$; down $a = g(\sin \theta - \mu_k \cos \theta)$; up $a = g(\sin \theta + \mu_k \cos \theta)$; rests if $\tan \theta \leq \mu_s$.
- **Blocks in contact** (F on m_1 , smooth): $a = \frac{F}{m_1 + m_2}$, contact force = $\frac{m_2 F}{m_1 + m_2}$ — the mass *ahead*.
- **Blocks joined by a string**, pulled by F : $a = \frac{F}{m_1 + m_2}$, $T = \frac{m_1 m_2}{m_1 + m_2} F$.

DERIVATIONS

$\vec{F} = d\vec{p}/dt \Rightarrow \vec{F} = m\vec{a}$; **impulse**

1. $\vec{p} = m\vec{v}$; second law in general form: $\vec{F} = \frac{d\vec{p}}{dt}$.
2. $\frac{d\vec{p}}{dt} = \frac{d(m\vec{v})}{dt} = m\frac{d\vec{v}}{dt} + \vec{v}\frac{dm}{dt}$.
3. **Constant mass** $\Rightarrow dm/dt = 0 \Rightarrow \vec{F} = m\frac{d\vec{v}}{dt} = m\vec{a}$.
4. Integrate over the contact time: $\int_{t_1}^{t_2} \vec{F} dt = \vec{p}_2 - \vec{p}_1$ — **impulse** $= \Delta\vec{p}$.

Same Δp over a longer $\Delta t \Rightarrow$ **smaller force**: bent knees on landing, airbags, drawing the hands back to catch a ball.

★ **Conservation of linear momentum from the 2nd + 3rd laws; gun recoil**

1. Isolated system: bodies A, B interact for a time Δt , no external force.
2. Third law: $\vec{F}_{AB} = -\vec{F}_{BA}$.
3. Second law on each: $\frac{d\vec{p}_A}{dt} = \vec{F}_{AB}$, $\frac{d\vec{p}_B}{dt} = \vec{F}_{BA}$.
4. Add: $\frac{d}{dt}(\vec{p}_A + \vec{p}_B) = \vec{F}_{AB} + \vec{F}_{BA} = 0$.
5. $\Rightarrow \vec{p}_A + \vec{p}_B$ constant: $m_1\vec{u}_1 + m_2\vec{u}_2 = m_1\vec{v}_1 + m_2\vec{v}_2$.
6. **Gun recoil**: gun (M) + bullet (m) at rest $\Rightarrow$ total $\vec{p} = 0$; after firing $0 = m\vec{v}_b + M\vec{v}_g$.
7. $\Rightarrow \vec{v}_g = -\frac{m}{M}\vec{v}_b$ — **opposite** to the bullet and small because $M \gg m$.

★ **Maximum speed on a level circular road**

1. Car of mass m , horizontal circular road of radius R . Forces: mg down, N up, static friction f — the **only** horizontal force.
2. Vertical equilibrium: $N = mg$.
3. Friction supplies F_c : $\frac{mv^2}{R} = f \leq \mu_s N = \mu_s mg$.
4. $\frac{v^2}{R} \leq \mu_s g \Rightarrow v \leq \sqrt{\mu_s Rg}$.

$v_{\max} = \sqrt{\mu_s Rg}$ — **independent of the car's mass; beyond it the car skids outward.**

★ **Banked road — optimum and maximum speed**

1. Road banked at θ , radius R . **Frictionless first**: forces are N ($\perp$ road) and mg .
 2. Resolve N : vertical $N \cos \theta = mg$; inward $N \sin \theta = \frac{mv_0^2}{R}$.
 3. Divide: $\tan \theta = \frac{v_0^2}{Rg} \Rightarrow v_0 = \sqrt{Rg \tan \theta}$ — the **optimum speed**, needing no friction.
 4. **With friction**, at $v_{\max}$ the car tends to slip *up* the slope, so $f = \mu_s N$ acts *down* it. Resolve vertically and horizontally:
 5. $N \cos \theta = mg + f \sin \theta$; $N \sin \theta + f \cos \theta = \frac{mv_{\max}^2}{R}$.
 6. Put $f = \mu_s N$; divide the second by the first to kill N and m :
 7. $\frac{v_{\max}^2}{Rg} = \frac{\sin \theta + \mu_s \cos \theta}{\cos \theta - \mu_s \sin \theta} = \frac{\tan \theta + \mu_s}{1 - \mu_s \tan \theta}$.
- $v_{\max} = \sqrt{Rg \frac{\mu_s + \tan \theta}{1 - \mu_s \tan \theta}}$. **Checks**: $\mu_s = 0 \Rightarrow \sqrt{Rg \tan \theta}$; $\theta = 0 \Rightarrow \sqrt{\mu_s Rg}$. **Friction reversed** $\Rightarrow v_{\min} = \sqrt{Rg \frac{\tan \theta - \mu_s}{1 + \mu_s \tan \theta}}$.

Atwood machine — acceleration and tension

1. $m_1 > m_2$ over a light frictionless pulley; inextensible string $\Rightarrow$ same $|a|$, same T both sides.
2. For m_1 (moving down): $m_1 g - T = m_1 a$.
3. For m_2 (moving up): $T - m_2 g = m_2 a$.
4. Add: $(m_1 - m_2)g = (m_1 + m_2)a \Rightarrow a = \frac{(m_1 - m_2)g}{m_1 + m_2}$.
5. Back-substitute: $T = m_2(g + a) = \frac{2m_1 m_2 g}{m_1 + m_2}$ (note $m_2 g < T < m_1 g$).

Traps — Ch 4

1. Action and reaction act on **different bodies** — never cancel, never both in one free-body diagram. N and mg on a resting block are *not* a pair.

2. On an incline $N = mg \cos \theta$, so friction is $\mu mg \cos \theta$ — not μmg .
3. $f_s = \mu_s N$ is the **maximum** only; below it, static friction equals the applied force (possibly zero).
4. In a lift the true weight mg never changes — only N does. In free fall $N = 0$, yet g still acts.
5. $\sqrt{Rg \tan \theta}$ is the **optimum** speed, not the maximum; with friction a band $v_{\min} \leq v \leq v_{\max}$ is allowed.

Ch 5 · Work, Energy and Power

FORMULAS

► Work

- $W = \vec{F} \cdot \vec{d} = Fd \cos \theta$ — scalar; θ between $\vec{F}$ and the **displacement [J]**
- $\theta < 90^\circ \Rightarrow +W$; $\theta = 90^\circ \Rightarrow$ **zero** (F_c , N on level ground); $\theta > 90^\circ \Rightarrow -W$ (friction; gravity on a rising body).
- Variable force: $W = \int_{x_1}^{x_2} F dx =$ **area under F - x** (below the axis $\Rightarrow$ negative).

► Energy

- $K = \frac{1}{2}mv^2 = \frac{p^2}{2m}$; $p = \sqrt{2mK}$ — kinetic energy **[J]**
- **Work-energy theorem**: $W_{\text{net}} = \Delta K = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$ — work of **all** forces
- $U = mgh$ — gravitational PE near the surface **[J]**
- $U = \frac{1}{2}kx^2$ — spring PE at extension/compression x ; k in N m^{-1} **[J]**
- $F = -\frac{dU}{dx}$ — negative slope of the PE curve; U minimum $\Rightarrow$ stable equilibrium
- $E = K + U =$ constant when only **conservative** forces do work.
- With friction: $\Delta K + \Delta U = W_{\text{non-cons}}$; heat $= f \times$ (relative sliding distance).

► Power

- $P_{\text{avg}} = \frac{W}{t}$; $P_{\text{inst}} = \frac{dW}{dt} = \vec{F} \cdot \vec{v} = Fv \cos \theta$ **[W = J s⁻¹]**
- 1 hp = 746 W; 1 kWh = 3.6×10^6 J (a unit of **energy**).

► Collisions

- Momentum is conserved in **every** collision; KE only in an **elastic** one.
- **1D elastic** (m_1, u_1 and m_2, u_2): $v_1 = \frac{m_1 - m_2}{m_1 + m_2}u_1 + \frac{2m_2}{m_1 + m_2}u_2$, $v_2 = \frac{2m_1}{m_1 + m_2}u_1 + \frac{m_2 - m_1}{m_1 + m_2}u_2$
- Elastic $\Leftrightarrow u_1 - u_2 = v_2 - v_1$ — approach speed = separation speed
- **Perfectly inelastic** ($u_2 = 0$): $v = \frac{m_1 u_1}{m_1 + m_2}$; $\Delta K_{\text{lost}} = \frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} u_1^2$; fraction lost $= \frac{m_2}{m_1 + m_2}$.
- $e = \frac{|v_2 - v_1|}{|u_1 - u_2|} = \frac{\text{separation}}{\text{approach}}$ — restitution coefficient; 1 elastic, 0 perfectly inelastic; dimensionless
- Ball dropped from h : $h' = e^2 h$; after n bounces $h_n = e^{2n} h$; $e = \sqrt{h'/h}$.
- **2D elastic** (m_2 at rest): $m_1 u_1 = m_1 v_1 \cos \theta_1 + m_2 v_2 \cos \theta_2$; $0 = m_1 v_1 \sin \theta_1 - m_2 v_2 \sin \theta_2$; $\frac{1}{2}m_1 u_1^2 = \frac{1}{2}m_1 v_1^2 + \frac{1}{2}m_2 v_2^2$ (3 equations, 4 unknowns).

► Vertical circle (string / track, radius R)

- Minimum speeds: top $\sqrt{gR}$, side (height R) $\sqrt{3gR}$, bottom $\sqrt{5gR}$.
- $T_{\text{bot}} - T_{\text{top}} = 6mg$ at **any** speed; at minimum $T_{\text{top}} = 0$, $T_{\text{bot}} = 6mg$, $T_{\text{side}} = 3mg$.
- At θ from the lowest point: $T = \frac{mv^2}{R} + mg \cos \theta$, $v^2 = v_{\text{bot}}^2 - 2gR(1 - \cos \theta)$.
- For a **rod** or a bead in a tube the minimum top speed is 0 (it can push).

DERIVATIONS

★ Work-energy theorem — constant and variable force

- Constant force:** from $v^2 = u^2 + 2as$, multiply by $\frac{1}{2}m$:
 $\frac{1}{2}mv^2 - \frac{1}{2}mu^2 = mas$.
- $ma = F$ and $Fs = W \Rightarrow W = \frac{1}{2}mv^2 - \frac{1}{2}mu^2 = \Delta K$.
- Variable force:** $W = \int_{x_1}^{x_2} F dx = \int ma dx = \int m \frac{dv}{dt} dx$.
- $\frac{dv}{dt} dx = \frac{dv}{dt} \frac{dx}{dt} dt = v dv$ (i.e. $a = v dv/dx$).
- $W = \int_u^v mv dv = \frac{1}{2}mv^2 - \frac{1}{2}mu^2 = \Delta K$.

$W_{\text{net}} = \Delta K$: **positive** $\Rightarrow$ speeds up, **negative** $\Rightarrow$ slows down, **zero** $\Rightarrow$ speed unchanged.

★ Potential energy of a spring, $U = \frac{1}{2}kx^2$

- Hooke's law: restoring force $F_s = -kx$ (x from the natural length; the sign means it opposes the displacement).
- Stretching slowly, the agent applies $F_{\text{ext}} = +kx$ over a further dx : $dW = kx dx$.
- $W_{\text{ext}} = \int_0^x kx dx = \frac{1}{2}kx^2$.
- Stored as elastic PE: $U(x) = \frac{1}{2}kx^2$ (the spring itself does $-\frac{1}{2}kx^2$).
- Equals the triangle under F - x : $\frac{1}{2} \cdot x \cdot kx$. U is **even** in x — equal stretch and compression store equal energy.

Work from x_1 to x_2 is $\frac{1}{2}k(x_2^2 - x_1^2)$ — not $\frac{1}{2}k(x_2 - x_1)^2$.

★ Conservation of mechanical energy — a freely falling body

- Mass m released from rest at height H ; ground as reference ($U = 0$); no air resistance, so only gravity acts.
- Top:** $v = 0 \Rightarrow K = 0$, $U = mgH$, $E = mgH$.
- Depth x below the top:** $v^2 = 2gx \Rightarrow K = mgx$, $U = mg(H - x)$, $E = mgH$.
- Ground:** $v^2 = 2gH \Rightarrow K = mgH$, $U = 0$, $E = mgH$.
- Same E at all three $\Rightarrow$ ME conserved; PE converts steadily into KE.

Halfway ($x = H/2$): $K = U = mgH/2$. **Ground speed $\sqrt{2gH}$, independent of m .**

★ 1D elastic collision — final velocities

- m_1, m_2 with u_1, u_2 along one line, final v_1, v_2 .
Momentum: $m_1(u_1 - v_1) = m_2(v_2 - u_2)$. (i)
- KE:** $m_1(u_1 - v_1)(u_1 + v_1) = m_2(v_2 - u_2)(v_2 + u_2)$. (ii)
- (ii) $\div$ (i): $u_1 + v_1 = u_2 + v_2$, i.e. $u_1 - u_2 = v_2 - v_1$ — **approach = separation**. (iii)
- Solve (i) and (iii) together for v_1, v_2 .

$$v_1 = \frac{m_1 - m_2}{m_1 + m_2} u_1 + \frac{2m_2}{m_1 + m_2} u_2, \quad v_2 = \frac{m_2 - m_1}{m_1 + m_2} u_2 + \frac{2m_1}{m_1 + m_2} u_1$$

Special case ($u_2 = 0$)	v_1	v_2	Physical picture
$m_1 = m_2$	0	u_1	velocities are exchanged (carrom, Newton's cradle)
$m_1 \gg m_2$ (heavy hits light)	$\approx u_1$	$\approx 2u_1$	heavy body ploughs on; light body shoots off at twice the speed
$m_1 \ll m_2$ (light hits heavy)	$\approx -u_1$	≈ 0	light body rebounds with almost the same speed; wall stays put

Perfectly inelastic collision — common velocity and KE loss; the 90° result

- m_1 (speed u_1) hits m_2 at rest and sticks: $m_1 u_1 = (m_1 + m_2)v \Rightarrow v = \frac{m_1 u_1}{m_1 + m_2}$.
- $K_i = \frac{1}{2}m_1 u_1^2$; $K_f = \frac{1}{2}(m_1 + m_2)v^2 = \frac{m_1^2 u_1^2}{2(m_1 + m_2)}$.
- $\Delta K = \frac{1}{2}m_1 u_1^2 \left[1 - \frac{m_1}{m_1 + m_2}\right] = \frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} u_1^2 > 0$ — heat, sound, deformation.
- 2D elastic, equal masses, target at rest:** $\vec{u} = \vec{v}_1 + \vec{v}_2$ and $u^2 = v_1^2 + v_2^2$.
- Square the vector equation: $u^2 = v_1^2 + v_2^2 + 2\vec{v}_1 \cdot \vec{v}_2 \Rightarrow \vec{v}_1 \cdot \vec{v}_2 = 0$.
- $\Rightarrow \vec{v}_1 \perp \vec{v}_2$: they separate at 90° (if both actually move).

★ Vertical circle — minimum speeds and $T_{\text{bot}} - T_{\text{top}} = 6mg$

- Mass m on a string of length R . **At the top** T_t and mg both point down, to the centre: $T_t + mg = \frac{mv_t^2}{R}$.
- A string only pulls $\Rightarrow$ critical case $T_t = 0$: $mg = \frac{mv_t^2}{R} \Rightarrow v_t^{\text{min}} = \sqrt{gR}$.
- Energy top $\rightarrow$ bottom** (drop $2R$, only gravity works):
 $\frac{1}{2}mv_b^2 = \frac{1}{2}mv_t^2 + mg(2R) \Rightarrow v_b^2 = v_t^2 + 4gR$.
- With $v_t^2 = gR$: $v_b^2 = 5gR \Rightarrow v_b^{\text{min}} = \sqrt{5gR}$.
- At the bottom** T_b is up (to the centre), mg down: $T_b - mg = \frac{mv_b^2}{R}$.
- Subtract: $T_b - T_t = \frac{m}{R}(v_b^2 - v_t^2) + 2mg = \frac{m}{R}(4gR) + 2mg$.

$T_b - T_t = 6mg$ **at any speed, since $v_b^2 - v_t^2 = 4gR$ always. At minimum:** $T_t = 0$, $T_b = 6mg$, $T_{\text{side}} = 3mg$, $v_{\text{side}} = \sqrt{3gR}$.

Traps — Ch 5

- In $W = Fd \cos \theta$, θ is between $\vec{F}$ and the **displacement**, not the horizontal. F_c and N (level ground) do **zero** work.
- The work-energy theorem uses the **net** work of all forces — gravity and friction included.
- Momentum is conserved in inelastic collisions too — only KE is lost. Never conserve KE there.
- Spring work $x_1 \rightarrow x_2$ is $\frac{1}{2}k(x_2^2 - x_1^2)$, **not** $\frac{1}{2}k(x_2 - x_1)^2$.
- Minimum speed at the top of a vertical circle is $\sqrt{gR}$, not zero (zero only for a rod or tube).

Ch 6 · System of Particles and Rotational Motion

FORMULAS

► Centre of mass

• $x_{\text{cm}} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$ — 2 particles; divides the join as $m_2 : m_1$ [m]

• n particles: $\vec{R}_{\text{cm}} = \frac{\sum m_i \vec{r}_i}{\sum m_i} = \frac{1}{M} \sum m_i \vec{r}_i$

• Continuous body: $\vec{R}_{\text{cm}} = \frac{1}{M} \int \vec{r} dm$

• Rod: CoM at $L/2$. Ring, disc, sphere, cylinder: geometric centre. Triangular lamina: centroid.

• $M\vec{v}_{\text{cm}} = \sum m_i \vec{v}_i = \vec{P}_{\text{total}}$; $M\vec{a}_{\text{cm}} = \vec{F}_{\text{ext}}$

► Torque, angular momentum

• $\vec{\tau} = \vec{r} \times \vec{F}$, $|\vec{\tau}| = rF \sin \theta = F \times (\perp \text{ distance from the axis})$ [N m]

• $\vec{L} = \vec{r} \times \vec{p}$, $|\vec{L}| = rp \sin \theta$; fixed axis: $L = I\omega$ [J s]

• $\vec{\tau}_{\text{ext}} = \frac{d\vec{L}}{dt}$; $\vec{\tau}_{\text{ext}} = 0 \Rightarrow \vec{L}$ constant $\Rightarrow I_1 \omega_1 = I_2 \omega_2$

• $\tau = I\alpha$ — rotational analogue of $F = ma$; $W = \tau\theta$; $P = \tau\omega$

► Moment of inertia

• $I = \sum m_i r_i^2 = \int r^2 dm$ — rotational inertia; $r = \perp$ **distance from the axis** [kg m²]

• $I = MK^2 \Rightarrow K = \sqrt{I/M}$ — radius of gyration: distance at which all the mass could sit [m]

$$\cdot K_{\text{rot}} = \frac{1}{2} I \omega^2 = \frac{L^2}{2I} \text{ [J]}$$

► Rotational kinematics (constant α)

$$\cdot \omega = \omega_0 + \alpha t; \theta = \omega_0 t + \frac{1}{2} \alpha t^2; \omega^2 = \omega_0^2 + 2\alpha\theta; \theta_{n^{\text{th}}} = \omega_0 + \frac{\alpha}{2}(2n-1)$$

$$\cdot \text{Links: } v = \omega r, a_t = \alpha r, a_c = \omega^2 r, s = r\theta.$$

► Equilibrium of a rigid body

$$\cdot \text{Both are needed: } \sum \vec{F} = 0 \text{ (no translation) and } \sum \vec{\tau} = 0 \text{ (no rotation).}$$

$$\cdot \text{Principle of moments: anticlockwise = clockwise moments, } F_1 d_1 = F_2 d_2; \text{MA} = \frac{\text{load}}{\text{effort}} = \frac{d_{\text{effort}}}{d_{\text{load}}}.$$

$$\cdot \text{Couple: equal, opposite, non-collinear forces. Net force 0, torque} = F \times d \text{ — the same about every point; gives pure rotation (tap, steering wheel).}$$

DERIVATIONS

$$\star \text{ Motion of the centre of mass: } M \vec{a}_{\text{cm}} = \vec{F}_{\text{ext}}$$

$$1. M \vec{R}_{\text{cm}} = \sum m_i \vec{r}_i, \text{ with } M = \sum m_i \text{ constant.}$$

$$2. \text{Differentiate: } M \vec{v}_{\text{cm}} = \sum m_i \vec{v}_i = \vec{P}_{\text{total}}.$$

$$3. \text{Again: } M \vec{a}_{\text{cm}} = \sum m_i \vec{a}_i = \sum \vec{F}_i.$$

$$4. \text{Split into internal and external: internal forces come in pairs } \vec{f}_{ij} = -\vec{f}_{ji} \text{ and cancel.}$$

$$5. \Rightarrow M \vec{a}_{\text{cm}} = \vec{F}_{\text{ext}}: \text{ the CoM moves like one particle of mass } M \text{ under } \vec{F}_{\text{ext}}.$$

$$6. \vec{F}_{\text{ext}} = 0 \Rightarrow \vec{a}_{\text{cm}} = 0 \Rightarrow \vec{v}_{\text{cm}} \text{ constant, } \vec{P}_{\text{total}} \text{ conserved.}$$

A shell exploding in flight: fragments scatter by internal forces, but the CoM keeps the original parabola.

Centre of mass of a uniform rod

$$1. \text{Rod of mass } M, \text{ length } L, \lambda = M/L; \text{ origin at one end, } x \text{ along the rod.}$$

$$2. \text{Element } dx \text{ at distance } x: dm = \lambda dx.$$

$$3. x_{\text{cm}} = \frac{1}{M} \int_0^L x \lambda dx = \frac{\lambda}{M} \cdot \frac{L^2}{2}.$$

$$4. \text{With } \lambda = M/L: x_{\text{cm}} = \frac{L}{2} \text{ — the midpoint, as symmetry demands.}$$

$$\star \vec{\tau} = d\vec{L}/dt \text{ and conservation of angular momentum}$$

$$1. \text{For a particle, } \vec{L} = \vec{r} \times \vec{p}.$$

$$2. \text{Product rule: } \frac{d\vec{L}}{dt} = \frac{d\vec{r}}{dt} \times \vec{p} + \vec{r} \times \frac{d\vec{p}}{dt}.$$

$$3. \text{First term: } \vec{v} \times m\vec{v} = 0 \text{ (parallel vectors).}$$

$$4. \text{Second term: } \frac{d\vec{p}}{dt} = \vec{F} \Rightarrow \vec{r} \times \vec{F} = \vec{\tau}.$$

$$5. \Rightarrow \vec{\tau} = \frac{d\vec{L}}{dt}. \text{ Over a system, internal torques cancel:}$$

$$\vec{\tau}_{\text{ext}} = \frac{d\vec{L}_{\text{tot}}}{dt}.$$

$$6. \vec{\tau}_{\text{ext}} = 0 \Rightarrow \vec{L} \text{ constant: } I_1 \omega_1 = I_2 \omega_2.$$

Skater / stool: arms in $\Rightarrow I$ down $\Rightarrow \omega$ up at fixed L ; $K_{\text{rot}} = L^2/2I$ rises, the extra energy being muscular work. Also a diver tucking.

STANDARD RESULTS

► Moments of inertia (M = mass, R = radius, L = length)

Body	Axis	I	K
Thin ring / hoop	centre, $\perp$ plane	MR^2	R
Thin ring	diameter	$\frac{1}{2}MR^2$	$R/\sqrt{2}$
Circular disc	centre, $\perp$ plane	$\frac{1}{2}MR^2$	$R/\sqrt{2}$
Circular disc	diameter	$\frac{1}{4}MR^2$	$R/2$
Thin rod	$\perp$, centre	$\frac{1}{12}ML^2$	$L/\sqrt{12}$
Thin rod	$\perp$, one end	$\frac{1}{3}ML^2$	$L/\sqrt{3}$
Solid sphere	diameter	$\frac{2}{5}MR^2$	$R\sqrt{2/5}$
Hollow sphere (shell)	diameter	$\frac{2}{3}MR^2$	$R\sqrt{2/3}$
Solid cylinder	own axis	$\frac{1}{2}MR^2$	$R/\sqrt{2}$
Hollow cylinder (thin)	own axis	MR^2	R
Solid cylinder	$\perp$, centre	$M\left(\frac{L^2}{12} + \frac{R^2}{4}\right)$	—

I grows as mass moves out: ring > shell > disc > solid sphere (same M, R).

► Linear $\leftrightarrow$ rotational analogy (fill either column from the other)

Linear motion	Rotational motion (fixed axis)
Displacement s	Angular displacement θ
Velocity $v = ds/dt$	Angular velocity $\omega = d\theta/dt$
Acceleration $a = dv/dt$	Angular acceleration $\alpha = d\omega/dt$
Mass m (inertia)	Moment of inertia I
Force F	Torque $\tau = rF \sin \theta$
$F = ma$	$\tau = I\alpha$
Momentum $p = mv$	Angular momentum $L = I\omega$
$F = dp/dt$	$\tau = dL/dt$
Impulse $F\Delta t = \Delta p$	Angular impulse $\tau\Delta t = \Delta L$
Work $W = Fs$	$W = \tau\theta$
KE = $\frac{1}{2}mv^2 = p^2/2m$	KE = $\frac{1}{2}I\omega^2 = L^2/2I$
Power $P = Fv$	$P = \tau\omega$
$v = u + at; s = ut + \frac{1}{2}at^2$	$\omega = \omega_0 + \alpha t; \theta = \omega_0 t + \frac{1}{2}\alpha t^2$
$v^2 = u^2 + 2as$	$\omega^2 = \omega_0^2 + 2\alpha\theta$

Only if covered in class — Axis theorems & rolling

$$\blacksquare \text{ Parallel axis: } I = I_{\text{cm}} + Md^2 \text{ (} d = \perp \text{ distance between parallel axes, one through the CoM). Check: rod about an end} = \frac{1}{12}ML^2 + M(L/2)^2 = \frac{1}{3}ML^2 \checkmark$$

$$\blacksquare \text{ Perpendicular axis (laminae only): } I_z = I_x + I_y, x, y \text{ in the plane, } z \perp, \text{ concurrent. Check: disc } \frac{1}{2}MR^2 = \frac{1}{4}MR^2 + \frac{1}{4}MR^2 \checkmark$$

$$\blacksquare \text{ Rolling without slipping: } v_{\text{cm}} = \omega R; \text{ contact point instantaneously at rest; top point at } 2v_{\text{cm}}.$$

$$\blacksquare \text{ Rolling KE: } K_{\text{total}} = \frac{1}{2}Mv_{\text{cm}}^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}Mv_{\text{cm}}^2 \left(1 + \frac{K^2}{R^2}\right); \frac{K_{\text{rot}}}{K_{\text{total}}} = \frac{K^2/R^2}{1+K^2/R^2} \text{ (} K \text{ here = radius of gyration).}$$

$$\blacksquare \text{ Down an incline } \theta: a = \frac{g \sin \theta}{1+K^2/R^2}; v_{\text{bot}} = \sqrt{\frac{2gh}{1+K^2/R^2}}; f = \frac{Mg \sin \theta (K^2/R^2)}{1+K^2/R^2}; \text{ needs } \mu_s \geq \frac{\tan \theta (K^2/R^2)}{1+K^2/R^2}.$$

$$\blacksquare \text{ Arrival order (smallest } K^2/R^2 \text{ first): sphere } \frac{2}{5} \rightarrow \text{disc } \frac{1}{2} \rightarrow \text{shell } \frac{2}{3} \rightarrow \text{ring } 1; \text{ independent of } M, R.$$

Traps — Ch 6

- I depends on the **axis**, not just the body. Always name the axis before quoting a value.
- In $I = \sum mr^2$, r is the $\perp$ **distance from the axis** — not from the centre or origin.
- L conserved and I down $\Rightarrow \omega$ up and $K_{\text{rot}} = L^2/2I$ up. KE is **not** conserved — internal work supplies it.
- Equilibrium needs **both** $\sum F = 0$ and $\sum \tau = 0$ — a couple has zero net force yet still rotates the body.
- $\vec{\omega}, \vec{\tau}, \vec{L}$ point **along the axis** (right-hand rule), not along the motion. $\vec{L} \parallel \vec{\omega}$ only about a principal axis.

Ch 7 · Gravitation (up to variation of g)

FORMULAS

► Kepler's laws

- 1. Orbits:** each planet moves in an **ellipse** with the Sun at one **focus** (circle = special case).
- 2. Areas:** the Sun–planet line sweeps **equal areas in equal times:** $\frac{dA}{dt} = \frac{L}{2m} = \text{const} \Rightarrow$ **fastest at perihelion**, slowest at aphelion.
- 3. Periods:** $T^2 \propto a^3$ (a = semi-major axis; $a = r$ for a circle): $T^2 = \frac{4\pi^2}{GM} r^3$

► Newton's law of gravitation

- $F = \frac{Gm_1m_2}{r^2}$ — attractive, along the line joining the masses; a 3rd-law pair **[N]**
- Vector form: $\vec{F}_{12} = -\frac{Gm_1m_2}{r^2} \hat{r}_{12}$ (the minus sign = attraction).
- $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$, $[G] = [\text{M}^{-1}\text{L}^3\text{T}^{-2}]$ — universal; independent of medium, temperature, pressure; Cavendish
- Superposition:** $\vec{F}_{\text{net}} = \vec{F}_1 + \vec{F}_2 + \dots$ — each pair acts independently.

- **Shell theorem:** (i) a uniform shell pulls an **outside** particle as if all its mass sat at the centre; (ii) **zero** force inside a shell $\Rightarrow$ at depth only the mass *below* pulls.

► Acceleration due to gravity

- $g = \frac{GM}{R^2} = \frac{4}{3}\pi G\rho R$ — surface value; independent of the falling body's mass [**m s⁻²**]
- $M = \frac{gR^2}{G} \approx 6 \times 10^{24}$ kg — mass of the Earth [**kg**]
- **Height:** $g_h = \frac{GM}{(R+h)^2} = \frac{gR^2}{(R+h)^2}$; for $h \ll R$: $g_h \approx g(1 - \frac{2h}{R})$
- **Depth:** $g_d = g(1 - \frac{d}{R})$; $g = 0$ at the centre ($d = R$)
- Weight $W = mg$ varies with location; **mass does not**.

DERIVATIONS

★ Kepler's second law from conservation of angular momentum

1. Planet of mass m at $\vec{r}$ from the Sun moves $d\vec{r} = \vec{v} dt$ in time dt .
2. It sweeps a thin triangle: $dA = \frac{1}{2}|\vec{r} \times d\vec{r}| = \frac{1}{2}|\vec{r} \times \vec{v}| dt$.
3. $\vec{L} = \vec{r} \times m\vec{v} \Rightarrow |\vec{r} \times \vec{v}| = L/m$.
4. $\Rightarrow$ areal velocity $\frac{dA}{dt} = \frac{L}{2m}$.
5. Gravity is **central** (along $\vec{r}$) $\Rightarrow \vec{\tau} = \vec{r} \times \vec{F} = 0 \Rightarrow \vec{L}$ constant.
6. m, L constant $\Rightarrow \frac{dA}{dt}$ constant: equal areas in equal times.

rv constant along the orbit $\Rightarrow$ small r (perihelion) forces large v . Kepler's 2nd law is L -conservation.

Kepler's third law for a circular orbit

1. Planet m circles a star M at radius r , period T , so $\omega = \frac{2\pi}{T}$.
2. Gravity supplies $F_c: \frac{GMm}{r^2} = m\omega^2 r$ — m cancels.
3. Substitute $\omega: \frac{GM}{r^2} = \frac{4\pi^2}{T^2}$.
4. $\Rightarrow T^2 = \frac{4\pi^2}{GM} r^3$, i.e. $T^2 \propto r^3$.

$4\pi^2/GM$ depends only on the central body — same for all planets of one Sun, all moons of one planet.

★ $g = GM/R^2$ and the mass of the Earth

1. Earth: mass M , radius R , uniform sphere $\Rightarrow$ by the shell theorem it pulls as if all its mass sat at the centre.
2. A body of mass m on the surface is at distance R :
 $F = \frac{GMm}{R^2}$.
3. Its weight is also $F = mg$.
4. Equate: $mg = \frac{GMm}{R^2} \Rightarrow g = \frac{GM}{R^2}$ — m cancels, so **all bodies fall alike**.
5. Invert: $M = \frac{gR^2}{G} = \frac{9.8 \times (6.4 \times 10^6)^2}{6.67 \times 10^{-11}} \approx 6 \times 10^{24}$ kg.
6. With $M = \frac{4}{3}\pi R^3 \rho$: $g = \frac{4}{3}\pi G\rho R$, so $g \propto \rho R$ — used to compare planets.

★ Variation of g with height

1. At height h the body is at distance $R + h$; the whole mass M still lies below it.
2. $g_h = \frac{GM}{(R+h)^2}$; divide by $g = \frac{GM}{R^2}$: $\frac{g_h}{g} = \frac{R^2}{(R+h)^2}$.
3. $\Rightarrow g_h = \frac{gR^2}{(R+h)^2} = g(1 + \frac{h}{R})^{-2}$ — exact, any h .
4. For $h \ll R$, binomial to first order: $(1 + \frac{h}{R})^{-2} \approx 1 - \frac{2h}{R}$.
5. $\Rightarrow g_h \approx g(1 - \frac{2h}{R})$; loss $\Delta g = \frac{2gh}{R}$, fractional $\frac{2h}{R}$.

★ Variation of g with depth

1. At depth d , $r = R - d$. Split the Earth (density ρ) into an **inner sphere** of radius $R - d$ and an **outer shell** of thickness d .
2. By the shell theorem the outer shell exerts **zero** force inside $\Rightarrow$ only the inner sphere counts.
3. $M' = \frac{4}{3}\pi(R - d)^3 \rho$; whole Earth $M = \frac{4}{3}\pi R^3 \rho$.
4. $g_d = \frac{GM'}{(R-d)^2} = \frac{G \cdot \frac{4}{3}\pi(R-d)^3 \rho}{(R-d)^2} = \frac{4}{3}\pi G\rho(R - d)$ — **linear** in $(R - d)$.
5. Surface: $g = \frac{4}{3}\pi G\rho R$; divide: $\frac{g_d}{g} = \frac{R-d}{R}$.
6. $\Rightarrow g_d = g(1 - \frac{d}{R})$; at the centre ($d = R$) $g = 0$.

Height vs depth, same small x : $g_h \approx g(1 - 2x/R)$ vs $g_d = g(1 - x/R) \Rightarrow g$ falls twice as fast going up as down. $g \rightarrow 0$ exactly at the centre, asymptotically at infinite height.

Only if covered in class — Variation of g with latitude

- At latitude λ a body circles with radius $R \cos \lambda$, so part of gravity is spent on F_c : $g' = g - \omega^2 R \cos^2 \lambda$.
- **Equator** ($\lambda = 0$): $g' = g - \omega^2 R$, minimum. **Poles** ($\lambda = 90^\circ$): $g' = g$, maximum (the pole lies on the axis).
- $\omega^2 R \approx 0.034 \text{ m s}^{-2}$ (0.35% of g). Stop the rotation and g rises everywhere but the poles; ω about $17\times$ larger $\Rightarrow$ weightless at the equator.
- The equatorial bulge ($R_{\text{eq}} > R_{\text{pole}}$) adds to this, so g is smallest at the equator.

Traps — Ch 7

1. $g_h = \frac{gR^2}{(R+h)^2}$ — keep the R^2 on top; $gR/(R+h)^2$ is dimensionally wrong.
2. **Height: inverse-square** ($\approx 1 - 2h/R$). **Depth: linear** ($1 - d/R$). Never put the factor 2 in the depth formula.
3. $g(1 - 2h/R)$ is binomial, valid only for $h \ll R$; for large h use the exact form.
4. $M = gR^2/G$, not gR/G . Check:
[LT⁻²][L²]/[M⁻¹L³T⁻²] = [M] ✓
5. G is universal; g is local and varies with height, depth, latitude and planet. g is not "the gravitational constant".

★ Night-before priority — do these eight in this order

1. **Projectile motion** — trajectory is a parabola, then T , H , R ; 45° maximum and complementary angles. Ch 3
2. **Banked road** — optimum $\sqrt{Rg \tan \theta}$, then the full v_{max} with friction. Ch 4
3. **1D elastic collision** — momentum + KE $\Rightarrow$ approach = separation $\Rightarrow v_1, v_2$ and the three special cases. Ch 5
4. **Work-energy theorem** — constant force from $v^2 = u^2 + 2as$, variable force by $\int mv dv$. Ch 5
5. **Vertical circle** — $\sqrt{gR}$ at the top, $\sqrt{5gR}$ at the bottom, $T_b - T_t = 6mg$. Ch 5
6. $\vec{\tau} = d\vec{L}/dt$ — differentiate $\vec{r} \times \vec{p}$, kill the $\vec{v} \times m\vec{v}$ term $\Rightarrow$ conservation of $\vec{L}$, skater. Ch 6
7. **Variation of g with height and depth** — $gR^2/(R+h)^2$ with the binomial step; inner-sphere argument for $g(1 - d/R)$. Ch 7
8. **Equations of motion** — graphical (v - t slope and area) and calculus (three integrations). Ch 2